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Build A Smart Touch-Controlled Multi-Doorbell System

Technology is advancing daily and everything is turning smarter. A number of traditional devices that we use in our homes are becoming smarter now, including doorbells that are evolving with new features and styles. Today you will find many fancy doorbells in the market with touch switch system, but these are very costly to set up and everybody cannot afford them.

So, we have decided to help you make your own touch display doorbell. This doorbell has a lot of unique features, which will give you a new experience altogether. An interesting feature of this doorbell is that it has separate bells for each room in your home or office. It also has a special notification system to let visitors know whether the office is closed or not, or if any family member is available or not.

The circuit diagram of the smart touch-controlled multi-doorbell system is shown in Fig. 2. It consists of Arduino Uno, 6.1cm (2.4-inch) TFT display, Bluetooth HC-05 and four piezo buzzers.

Setting up Arduino library

Install the required library in Arduino IDE. For this, go to Sketch menu and click on Manage Library. Search for library names—Adafruit GFX and TouchScreen—and click on Install.

Every touchscreen has its own library. Hence, find a library based on your TFT screen model, and install it. Here, we have used MCUFriend 2.4 TFT LCD library St7789v and installed it in Arduino IDE. Refer Figs 3, 4 and 5 for details.

Coding Arduino

Initialise the libraries in the code and declare the required variables (for example, home1, home2, home3 and home4) that you want to control for separate rooms or office cabins. Then,

add fonts and colour codes for the TFT display to the following code:

```
#include <Fonts/FreeMonoBold24pt7b.h>
#include <Fonts/FreeMonoBold9pt7b.h>
#include <Fonts/FreeSerif12pt7b.h>
#include <FreeDefaultFonts.h>
//font
```

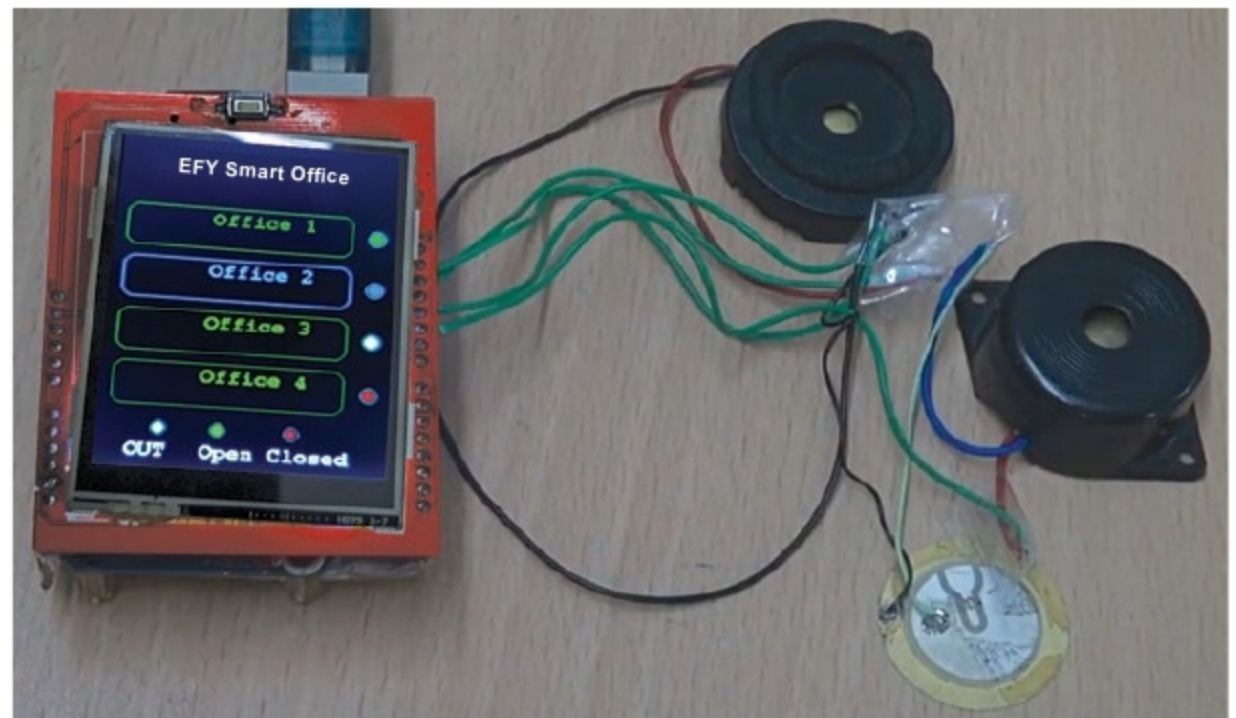


Fig. 1: Author's prototype

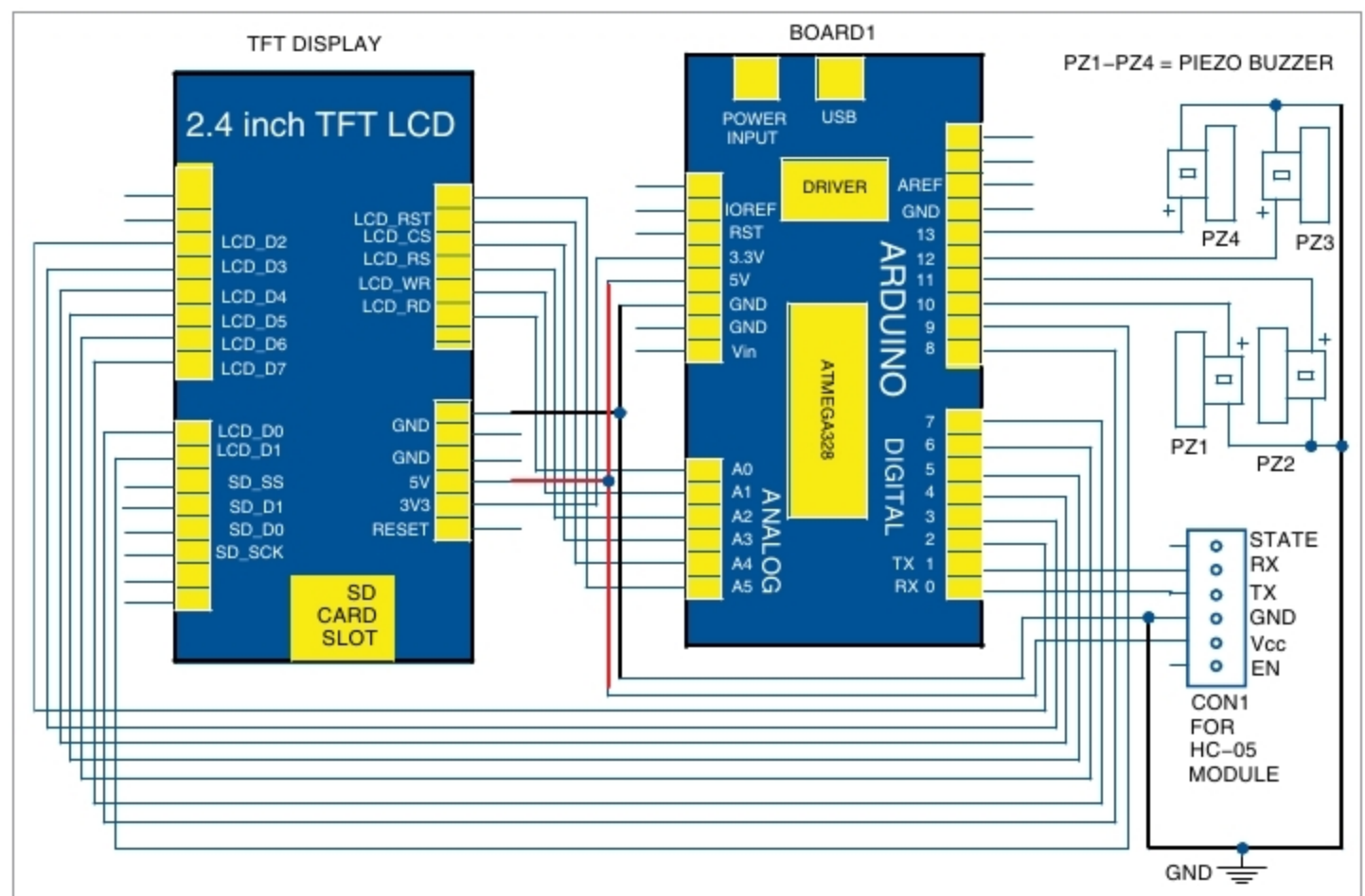


Fig. 2: Circuit diagram of the smart touch-controlled multi-doorbell system